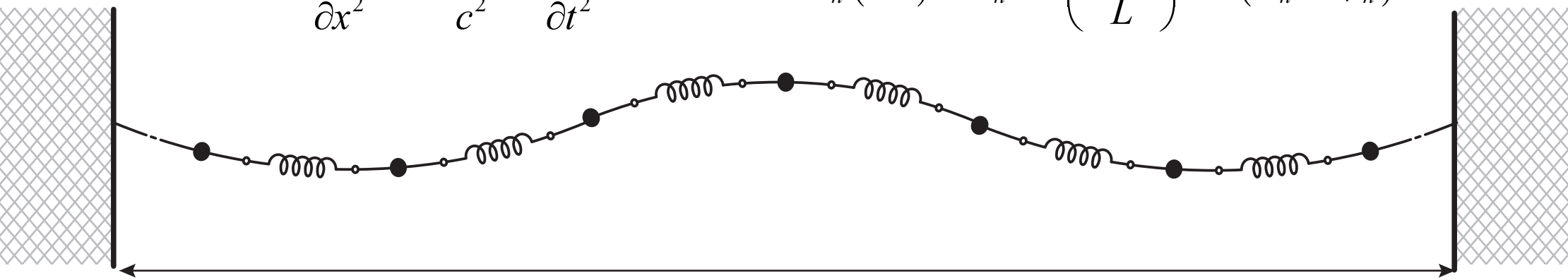
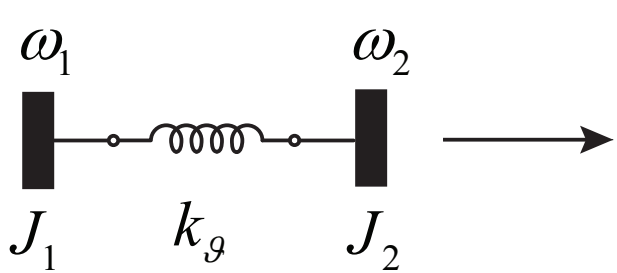
one dominant linear dispacement harmonic

$$\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u(x,t)}{\partial t^2} \longrightarrow u_n(x,t) = \alpha_n \sin\left(\frac{n\pi x}{L}\right)\cos\left(\omega_n t + \varphi_n\right)$$



isomorphism (from PDEs to ODEs)

One dominant torsional displacement harmonic



$$\begin{cases} J_1\dot{\omega}_1 = -\tau_{12} \\ J_2\dot{\omega}_2 = \tau_{12} \\ \dot{\tau}_{12} = k_\vartheta\left(\omega_1 - \omega_2\right) \end{cases}$$